

- Description

This module manages the cooling system of a smart beverage vending machine, ensuring that:

1. The system continuously monitors the current refrigeration temperature (in °C).
2. If the temperature exceeds a predefined maximum threshold, the cooling system is activated to lower the temperature.
3. If the temperature falls below a predefined minimum threshold, the cooling system is deactivated to prevent freezing.

This system ensures that beverages are stored at an optimal serving temperature.

- Specification

module SmartVendingCoolingControl

var currentTemperature, minTemperature, maxTemperature: real

var coolingActive: boolean

process updateTemperature(newTemperature: real) returns (currentTemperature: real, coolingActive: boolean)

functional scenario form:

currentTemperature > maxTemperature && coolingActive = true, output "ALERT: Temperature too high (" + currentTemperature + "°C). Cooling system activated.", Cooling system activated.

||

currentTemperature < minTemperature && coolingActive = false, output "ALERT: Temperature too low (" + currentTemperature + "°C). Cooling system deactivated.", Cooling system deactivated.

||

currentTemperature <= maxTemperature && output "Temperature optimal (" + currentTemperature + "°C). No action needed.", No control action required.

end process

end module

- Java Code

```
package code;
public class c95_SmartVendingMachineCooling {
    private double minTemperature;
    private double maxTemperature;
    private double currentTemperature;
    private boolean coolingActive;
    public c95_SmartVendingMachineCooling(double minTemperature, double
maxTemperature, double initialTemperature) {
        this.minTemperature = minTemperature;
        this.maxTemperature = maxTemperature;
        this.currentTemperature = initialTemperature;
        this.coolingActive = false;
    }
}
```

```

    }
    public void updateTemperature(double newTemperature) {
        currentTemperature = newTemperature;
        adjustCoolingSystem();
        System.out.printf("Current Temperature: %.2f°C, Cooling: %s\n",
            currentTemperature, coolingActive ? "ON" : "OFF");
        if (coolingActive) {
            System.out.println("Cooling system activated to reduce temperature.");
        } else {
            System.out.println("Temperature is within the optimal range.");
        }
    }
}

private void adjustCoolingSystem() {
    if (currentTemperature > maxTemperature) {
        coolingActive = true;
    } else if (currentTemperature < minTemperature) {
        coolingActive = false;
    }
}

public double getCurrentTemperature() {
    return currentTemperature;
}

public static void main(String[] args) {
    c95_SmartVendingMachineCooling vendingCooling = new c95_SmartVendingMachineCooling(2.0, 8.0, 5.0);

    // testcase-VT:
    //     vendingCooling.updateTemperature(10.0);
    //     vendingCooling.updateTemperature(5.0);
    //     vendingCooling.updateTemperature(1.0);

    // testcase-FT:
    //     vendingCooling.updateTemperature(-1.7976931348623157E308);
    //     vendingCooling.updateTemperature(-4.9E-324);
    //     vendingCooling.updateTemperature(1.7292085181828155E293);

    // testcase-Z3:
    //     vendingCooling.updateTemperature(10.0);
    //     vendingCooling.updateTemperature(5.0);
    //     vendingCooling.updateTemperature(1.0);

    // testcase-UVT:
    vendingCooling.updateTemperature(10.0);

```

```

        vendingCooling.updateTemperature(5.0);
        vendingCooling.updateTemperature(1.0);
    }
}

```

- Coverage report

1. Test case number: 3
2. V-Method testing branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	1.0 %	8	834	842
▼ src	1.0 %	8	834	842
▼ code	1.0 %	8	834	842
▼ c95_SmartVendingMachineCooling.jar	100.0 %	8	0	8
▼ c95_SmartVendingMachineCooling	100.0 %	8	0	8
main(String[])		0	0	0
c95_SmartVendingMachineCool		0	0	0
adjustCoolingSystem()	100.0 %	4	0	4
updateTemperature(double)	100.0 %	4	0	4

3. Vibration testing instruction coverage: 96.7%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.4 %	89	21,631	21,720
▼ src	0.4 %	89	21,631	21,720
▼ code	0.4 %	89	21,631	21,720
▼ c95_SmartVendingMachineCooling.jar	96.7 %	89	3	92
▼ c95_SmartVendingMachineCooling	96.7 %	89	3	92
main(String[])	100.0 %	17	0	17
c95_SmartVendingMachineCool	100.0 %	15	0	15
adjustCoolingSystem()	100.0 %	20	0	20
updateTemperature(double)	100.0 %	37	0	37

4. Fuzz testing branch coverage: 87.5%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.8 %	7	835	842
▼ src	0.8 %	7	835	842
▼ code	0.8 %	7	835	842
▼ c95_SmartVendingMachineCooling.jar	87.5 %	7	1	8
▼ c95_SmartVendingMachineCooling	87.5 %	7	1	8
main(String[])		0	0	0
c95_SmartVendingMachineCool		0	0	0
adjustCoolingSystem()	75.0 %	3	1	4
updateTemperature(double)	100.0 %	4	0	4

5. Fuzz testing instruction coverage: 96.7%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.4 %	89	21,631	21,720
▼ src	0.4 %	89	21,631	21,720
▼ code	0.4 %	89	21,631	21,720
▼ c95_SmartVendingMachineCooling.jar	96.7 %	89	3	92
▼ c95_SmartVendingMachineCooling	96.7 %	89	3	92
main(String[])	100.0 %	17	0	17
c95_SmartVendingMachineCool	100.0 %	15	0	15
adjustCoolingSystem()	100.0 %	20	0	20
updateTemperature(double)	100.0 %	37	0	37

6. Z3 branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	1.0 %	8	834	842
▼ src	1.0 %	8	834	842
▼ code	1.0 %	8	834	842
▼ c95_SmartVendingMachineCooling.java	100.0 %	8	0	8
▼ c95_SmartVendingMachineCooling	100.0 %	8	0	8
main(String[])		0	0	0
c95_SmartVendingMachineCool		0	0	0
adjustCoolingSystem()	100.0 %	4	0	4
updateTemperature(double)	100.0 %	4	0	4

7. Z3 instruction coverage: 96.7%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.4 %	89	22,940	23,029
▼ src	0.4 %	89	22,940	23,029
▼ code	0.4 %	89	22,940	23,029
▼ c95_SmartVendingMachineCooling.java	96.7 %	89	3	92
▼ c95_SmartVendingMachineCooling	96.7 %	89	3	92
main(String[])	100.0 %	17	0	17
c95_SmartVendingMachineCooling(c	100.0 %	15	0	15
adjustCoolingSystem()	100.0 %	20	0	20
updateTemperature(double)	100.0 %	37	0	37

8. Unimproved V-method branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	1.0 %	8	834	842
▼ src	1.0 %	8	834	842
▼ code	1.0 %	8	834	842
▼ c95_SmartVendingMachineCooling.java	100.0 %	8	0	8
▼ c95_SmartVendingMachineCooling	100.0 %	8	0	8
main(String[])		0	0	0
c95_SmartVendingMachineCool		0	0	0
adjustCoolingSystem()	100.0 %	4	0	4
updateTemperature(double)	100.0 %	4	0	4

9. Unimproved V-method instruction coverage: 96.7%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.4 %	89	22,940	23,029
▼ src	0.4 %	89	22,940	23,029
▼ code	0.4 %	89	22,940	23,029
▼ c95_SmartVendingMachineCooling.java	96.7 %	89	3	92
▼ c95_SmartVendingMachineCooling	96.7 %	89	3	92
main(String[])	100.0 %	17	0	17
c95_SmartVendingMachineCooling(c	100.0 %	15	0	15
adjustCoolingSystem()	100.0 %	20	0	20
updateTemperature(double)	100.0 %	37	0	37